



MRC Cognition
and Brain
Sciences Unit



UNIVERSITY OF
CAMBRIDGE

EEG/MEG 2:

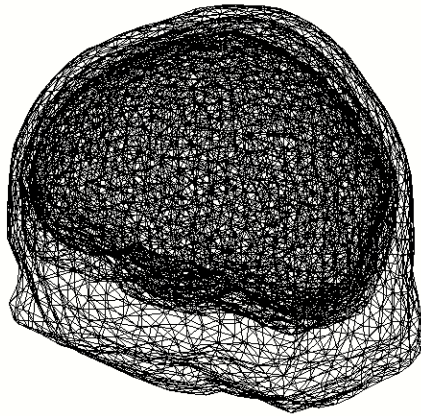
(Linear) Source Estimation

Olaf Hauk

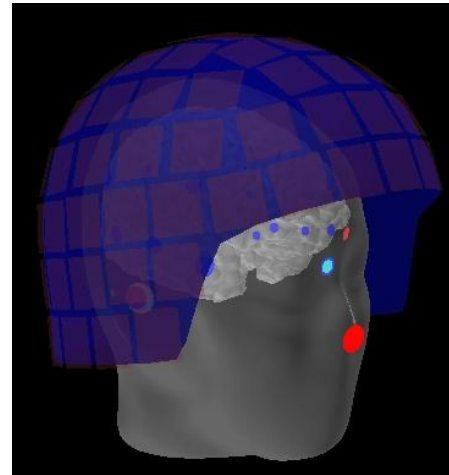
olaf.hauk@mrc-cbu.cam.ac.uk

Ingredients for Source Estimation

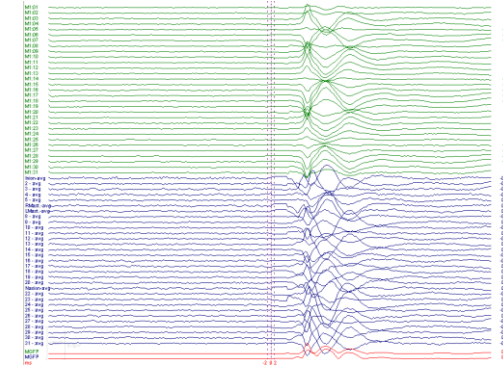
Volume Conductor/
Head Model



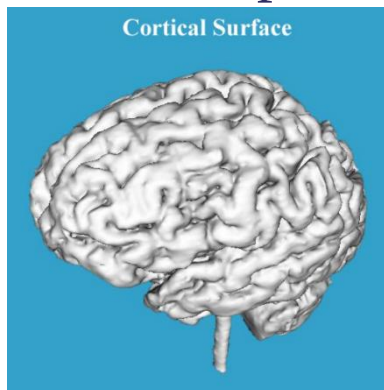
Coordinate
Transformation



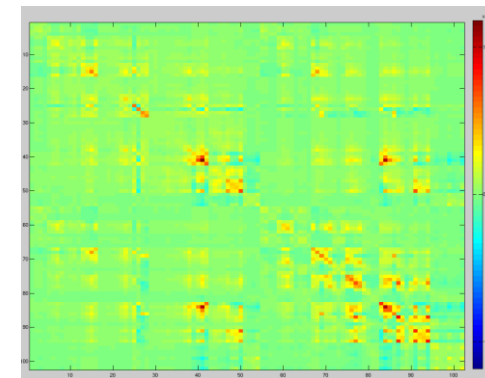
MEG data



Source Space

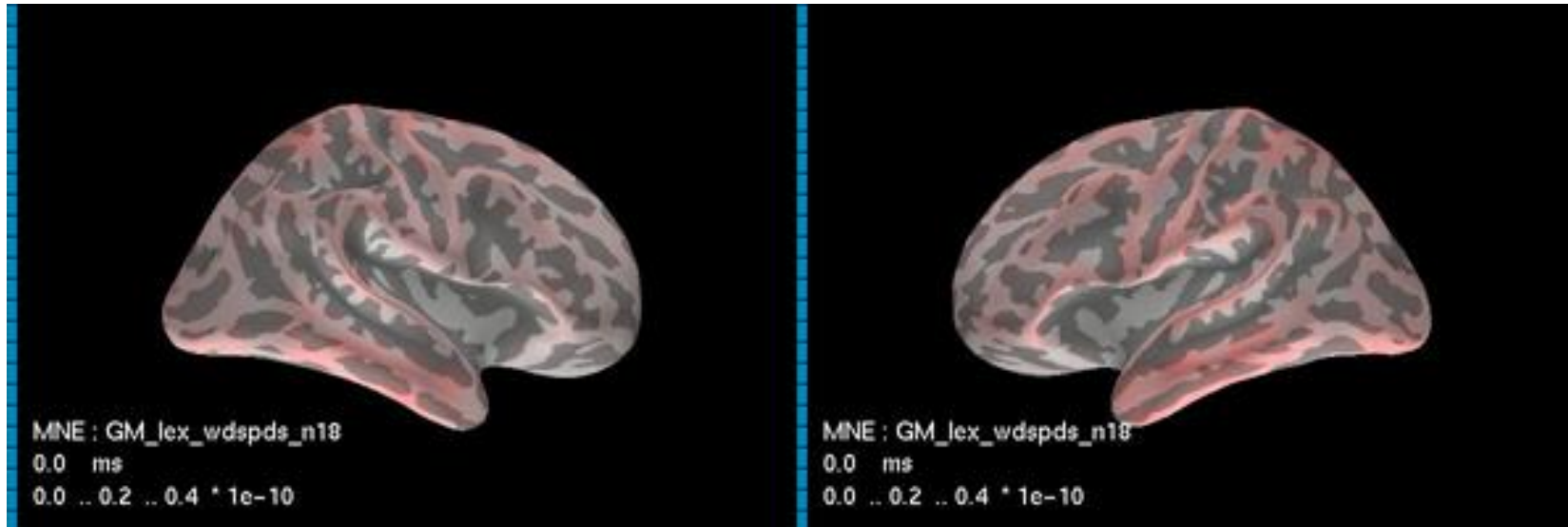


Noise/Covariance Matrix



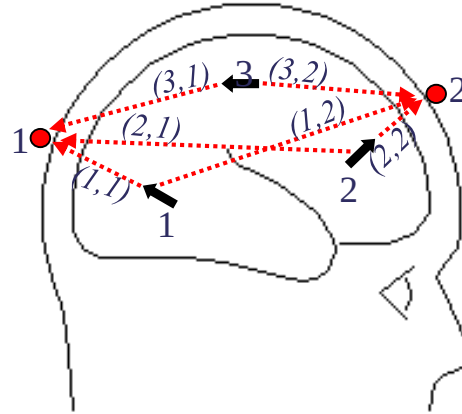
Our Goal: Spatio-Temporal Brain Dynamics

“Brain Movies”



The EEG/MEG Inverse Problem

The EEG/MEG Forward Problem



$$\begin{array}{c}
 \text{data} \quad \text{"leadfield"} \quad \text{dipoles} \\
 \begin{matrix} 1 \\ 2 \end{matrix} \cdot \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} = \begin{pmatrix} 0.5 & 0 & 0.3 \\ 0 & 1 & -0.3 \end{pmatrix} \begin{pmatrix} j_1 \\ j_2 \\ j_3 \end{pmatrix} \begin{matrix} \nwarrow 1 \\ \nearrow 2 \\ \nwarrow 3 \end{matrix} \\
 \text{?} \\
 \text{inversion} \\
 \begin{matrix} \nwarrow 1 \\ \nearrow 2 \\ \nwarrow 3 \end{matrix} \begin{pmatrix} j_1 \\ j_2 \\ j_3 \end{pmatrix} = \begin{pmatrix} 1.5034 & 0.1241 \\ 0.2483 & 0.9379 \\ 0.8276 & -0.2069 \end{pmatrix} * \begin{pmatrix} d_1 \\ d_2 \end{pmatrix} \begin{matrix} 1 \\ 2 \end{matrix} \cdot
 \end{array}$$

$$j_1 + j_2 = 1$$

under-determined problem, no unique solution

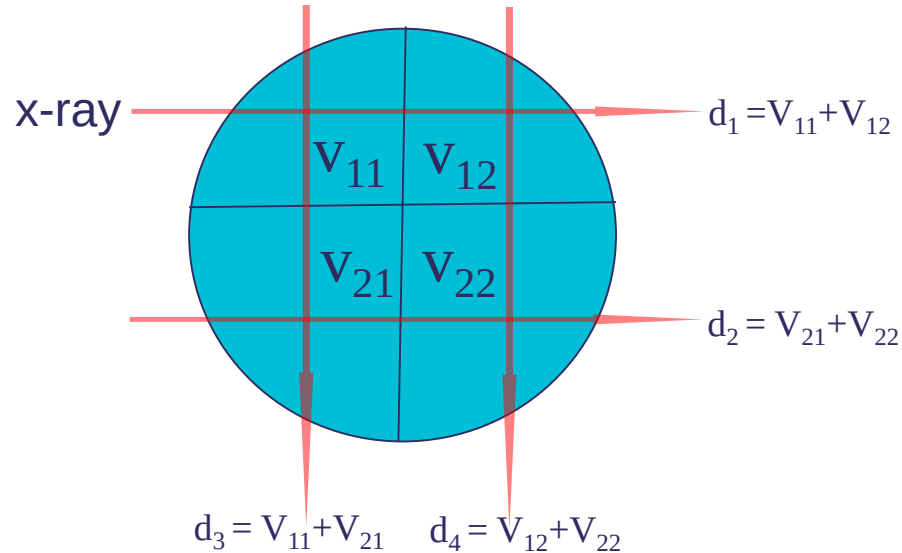
$$\mathbf{d} = \mathbf{L}\mathbf{j}$$

d: data (n_sensors x 1) **L**: "leadfield" (n_sensors x n_dipoles), **j**: dipoles (n_dipoles x 1)

Usually n_dipoles >> n_sensors.

EEG/MEG “Scanning” is not “Tomography”

Tomography (CT, fMRI...)



$$d_1 = V_{11} + V_{12}$$

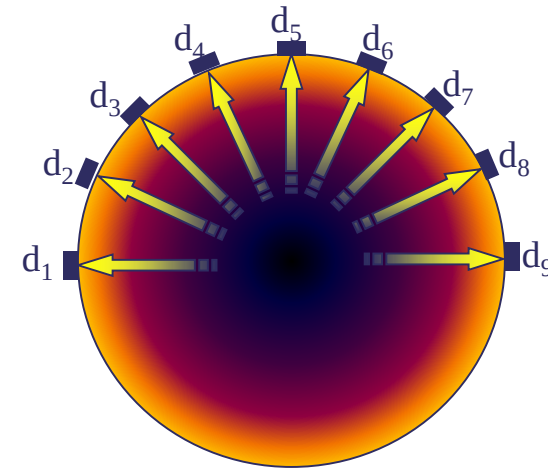
$$d_2 = V_{21} + V_{22}$$

$$d_3 = V_{11} + V_{21}$$

$$d_4 = V_{12} + V_{22}$$

Available information is determined by
the equipment/experimenter

EEG/MEG



$$d_1 = V_{11} + V_{12} + V_{13} + V_{14} \dots$$

$$d_2 = V_{21} + V_{22} + V_{23} + V_{24} \dots$$

Information is lost during
measurement

Cannot be retrieved by
mathematics

Inherently limits spatial resolution

Why Inverse “Problem”?

Without additional constraints the solution is non-unique, i.e. there are infinitely many solutions

What is the solution to

$$x_1 + x_2 = 1$$

Maybe

$$x_1 = 0 ; x_2 = 1 \quad ?$$

$$x_1 = 1 ; x_2 = 0 \quad ?$$

$$x_1 = 1000 ; x_2 = -999 \quad ?$$

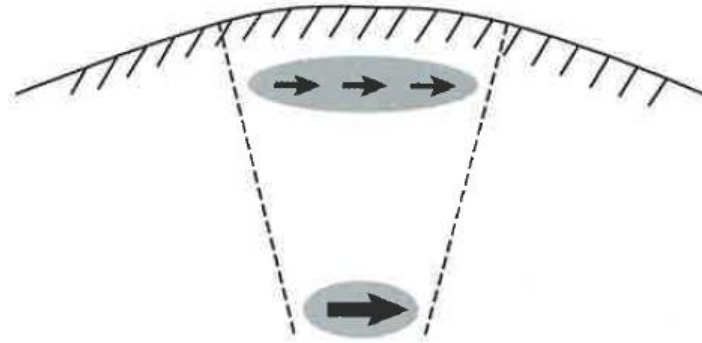
$$x_1 = \pi ; x_2 = (1-\pi) \quad ?$$

The “minimum norm solution” is:

$$x_1 = 0.5 ; x_2 = 0.5$$

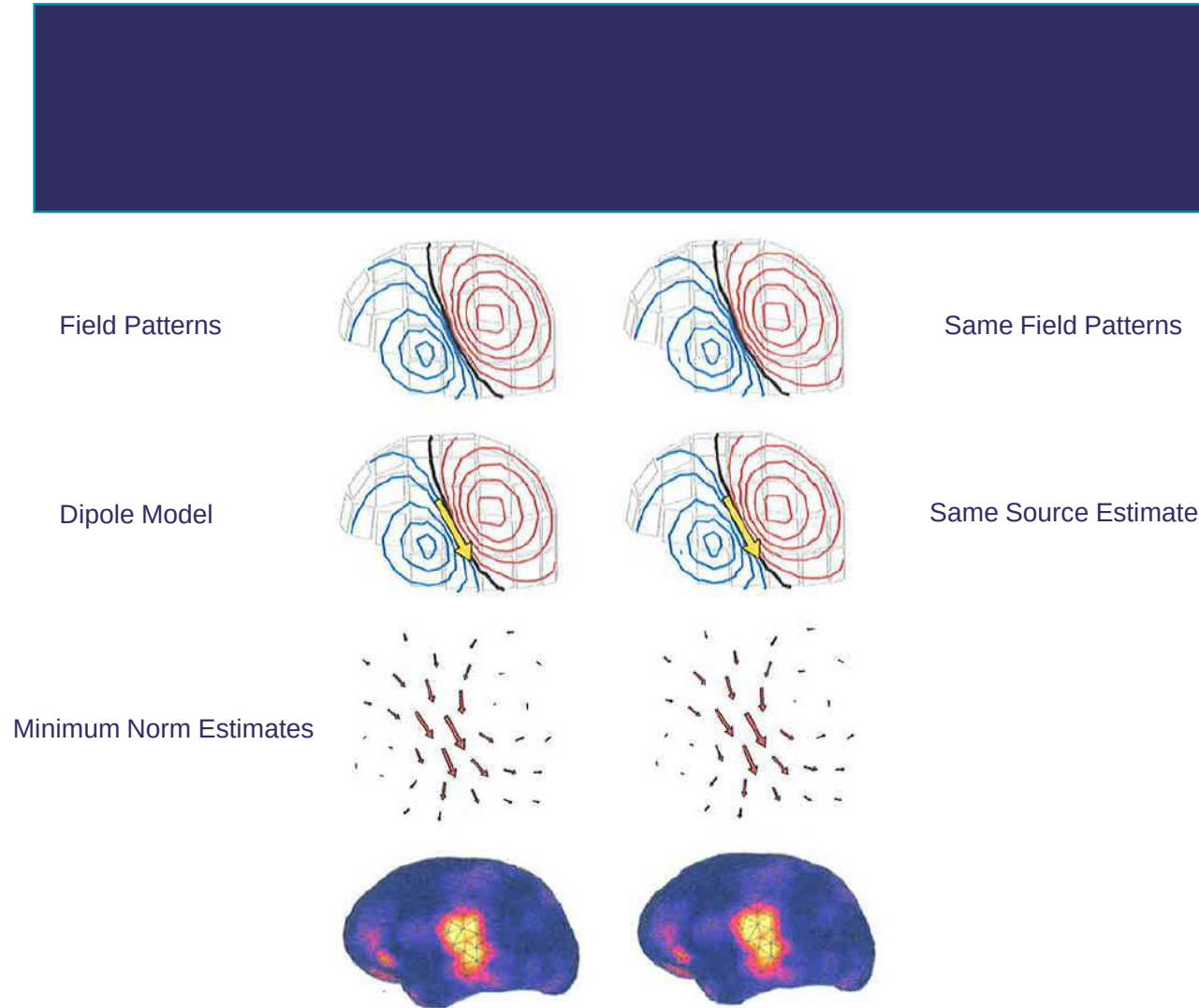
with $(0.5^2 + 0.5^2)=0.5$ the minimum norm among all possible solutions.

Examples for Non-Uniqueness



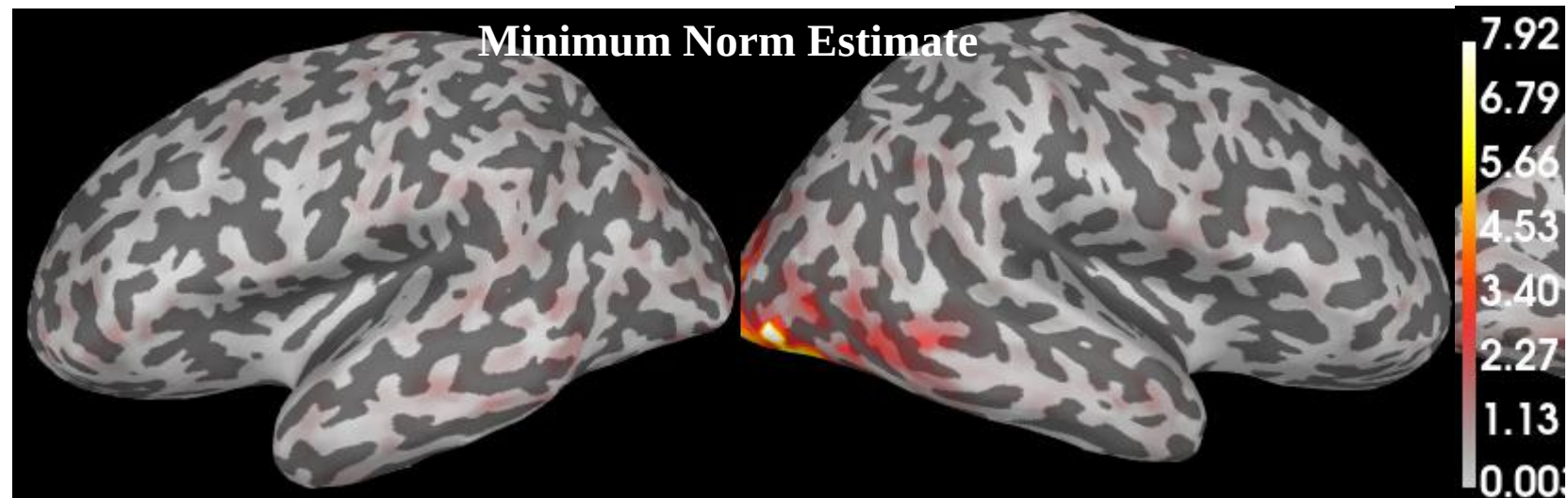
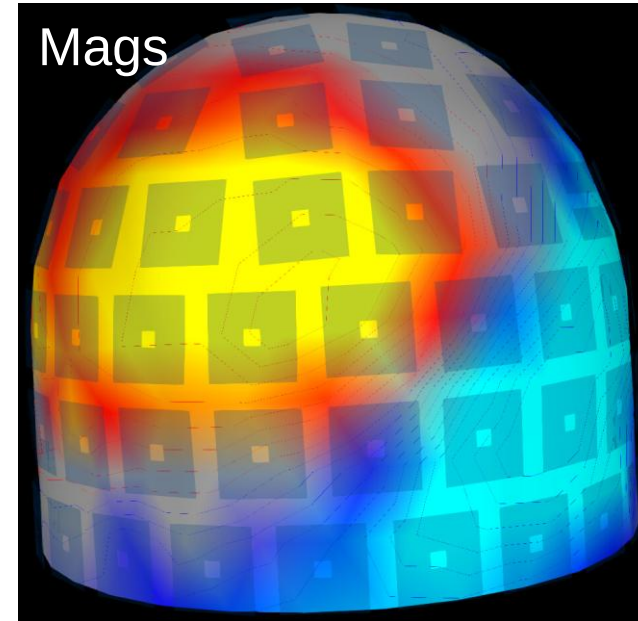
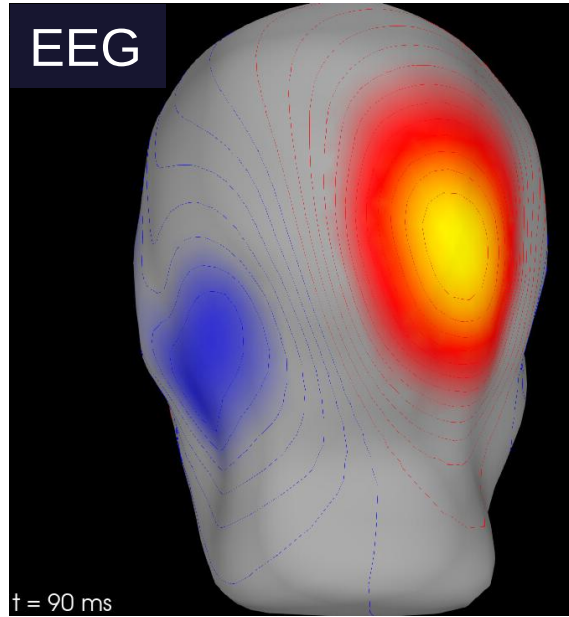
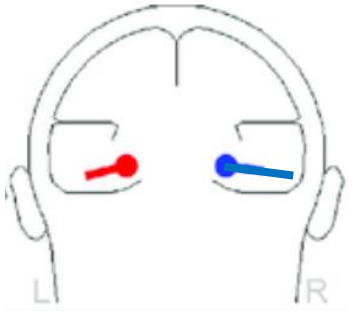
A distributed superficial distribution may be indistinguishable from a focal deep source.

Examples for Non-Uniqueness



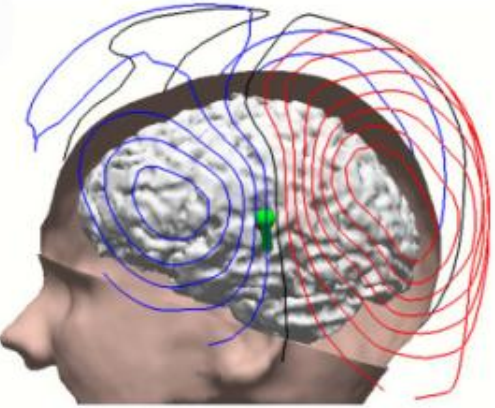
Example: Visually Evoked Activity ~100 ms

Checkerboard to
left visual field

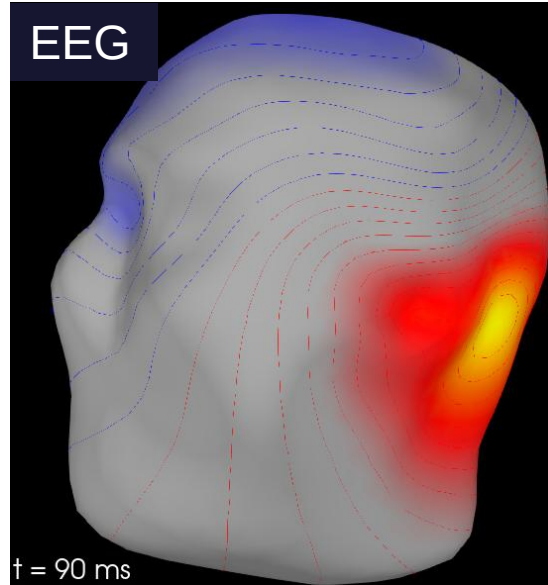


Example: Auditorily Evoked Activity

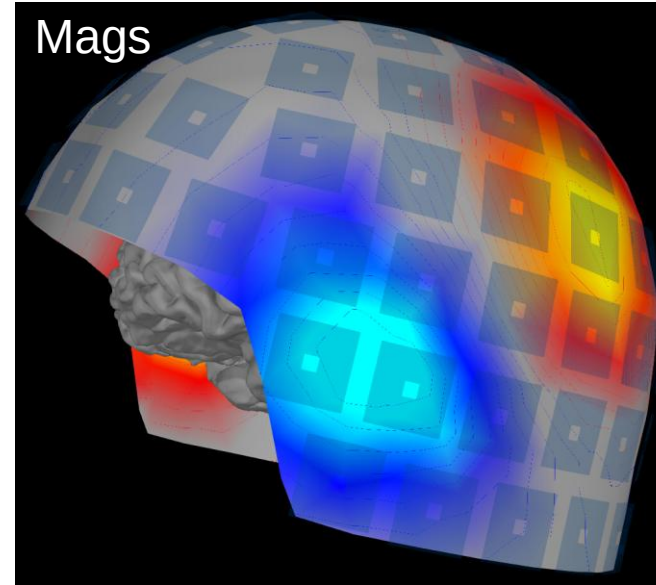
Tone to right ear



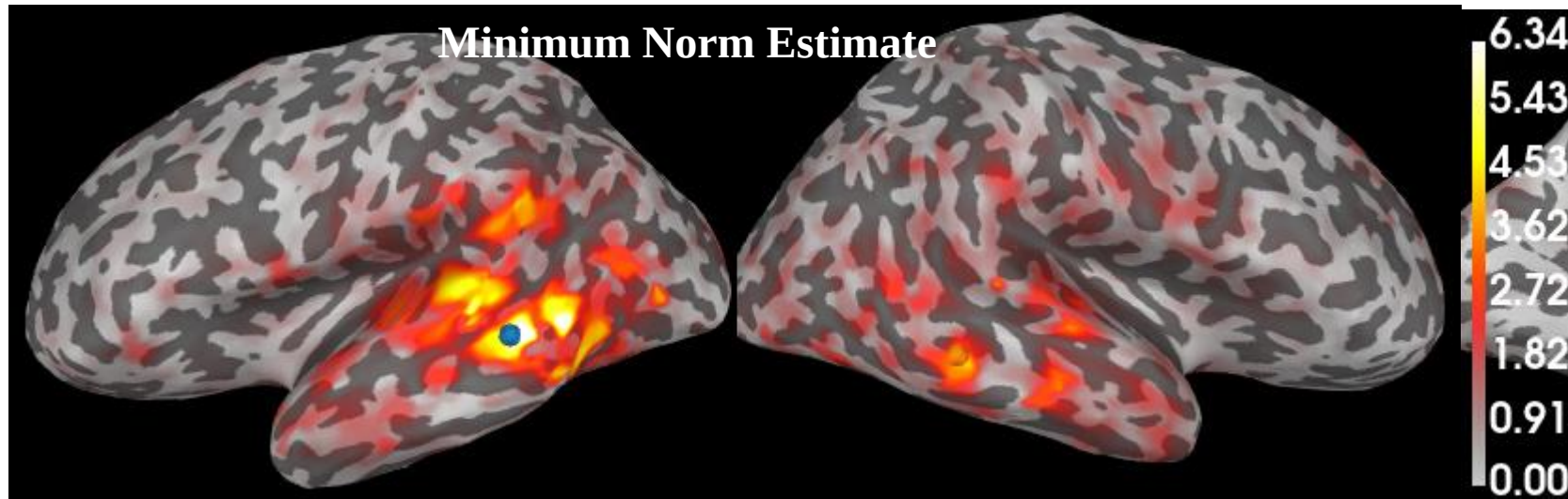
EEG



Mags



Minimum Norm Estimate

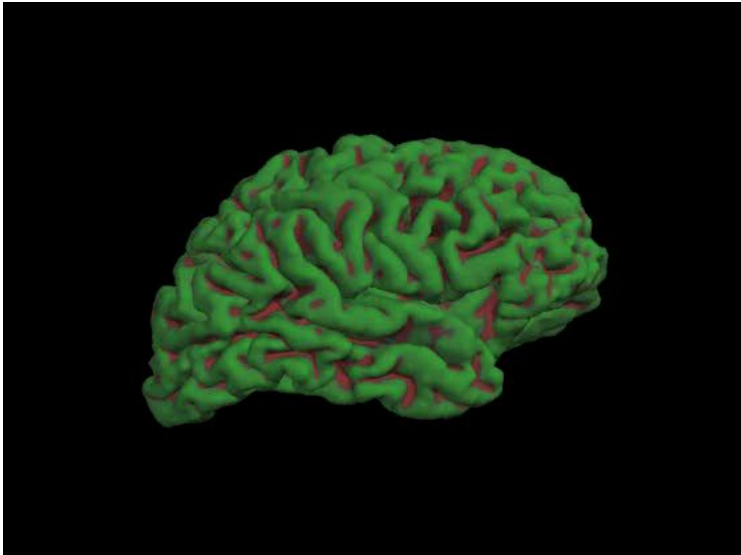


The Forward Problem and Head Modelling

Source Space and Head Model

Source Space

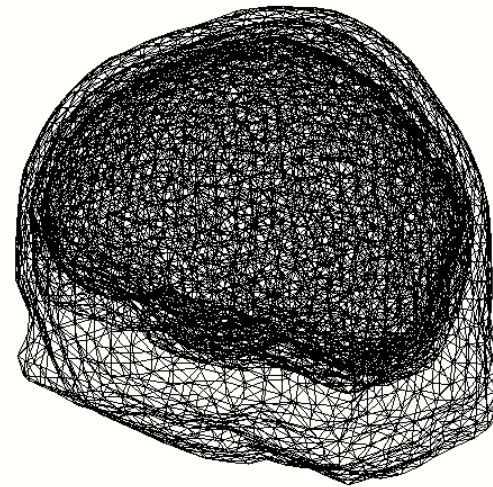
Where active sources may be located,
e.g. grey matter, 3D volume



<http://www.cogsci.ucsd.edu/~sereno/movies.html>

Volume Conductor/Head Model

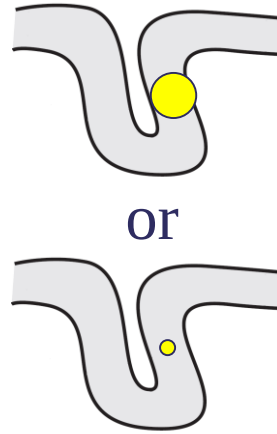
How we model conductivities/currents/potentials/fields in the head
e.g. sphere or realistic 1- or 3-compartments from MRI



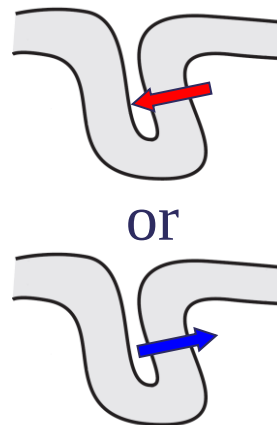
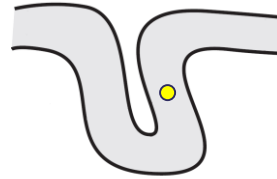
Sometimes “standard head models” are used, when no individual MRIs available.

SPM uses the same “canonical mesh” as source space for every subjects, but adjusts it individually.

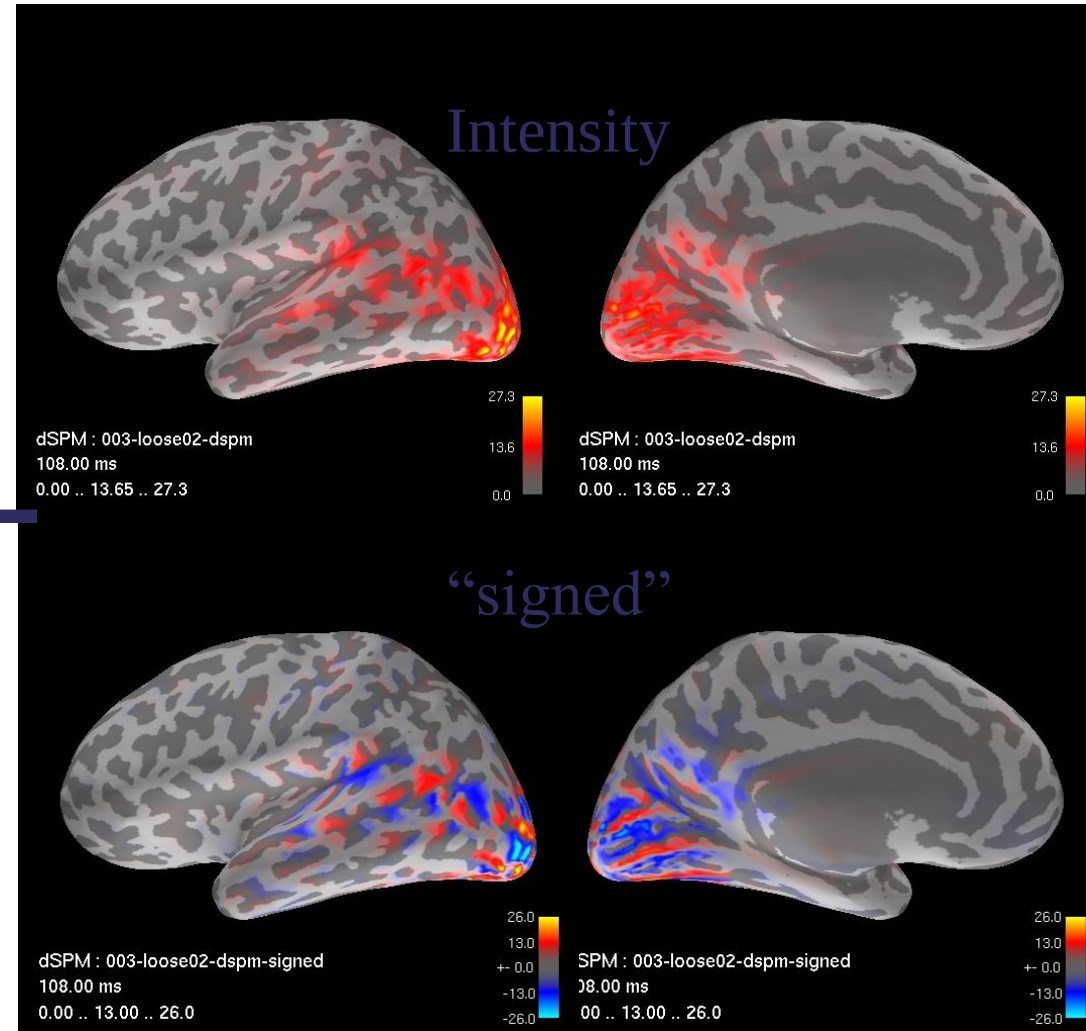
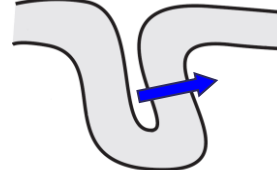
Direction of Current Flow



or



or

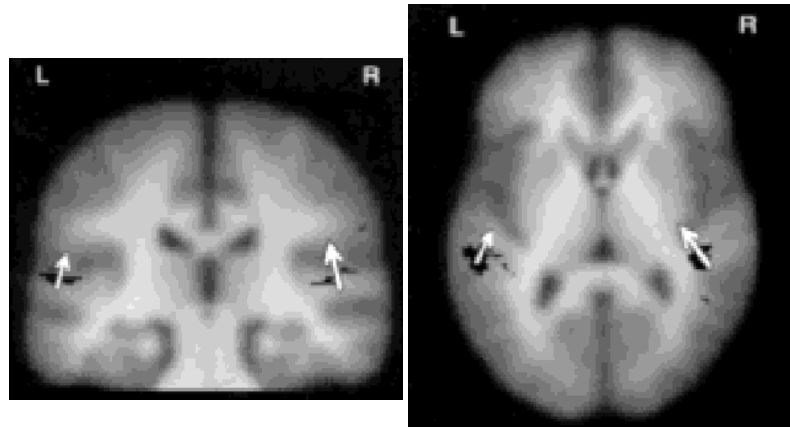


Solutions To The Inverse Problem – Source Estimation

Paths To Uniqueness

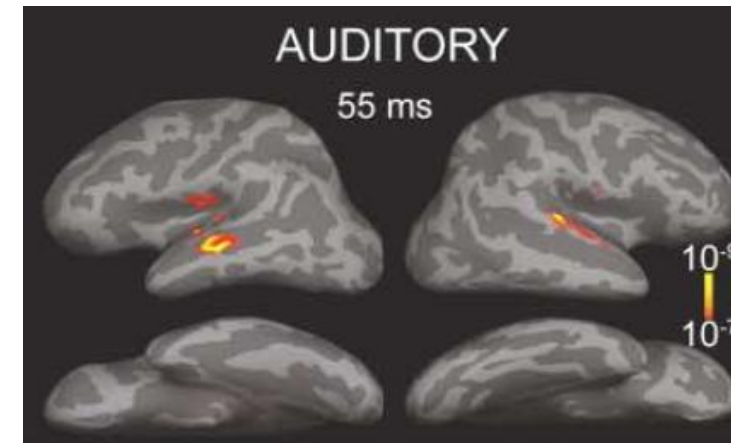
Dipole Fitting/Scanning

1. Assume there are only a few distinct sources
2. Iteratively adjust the location, orientation and strength of a few dipoles...
3. ...until the result best fits the data



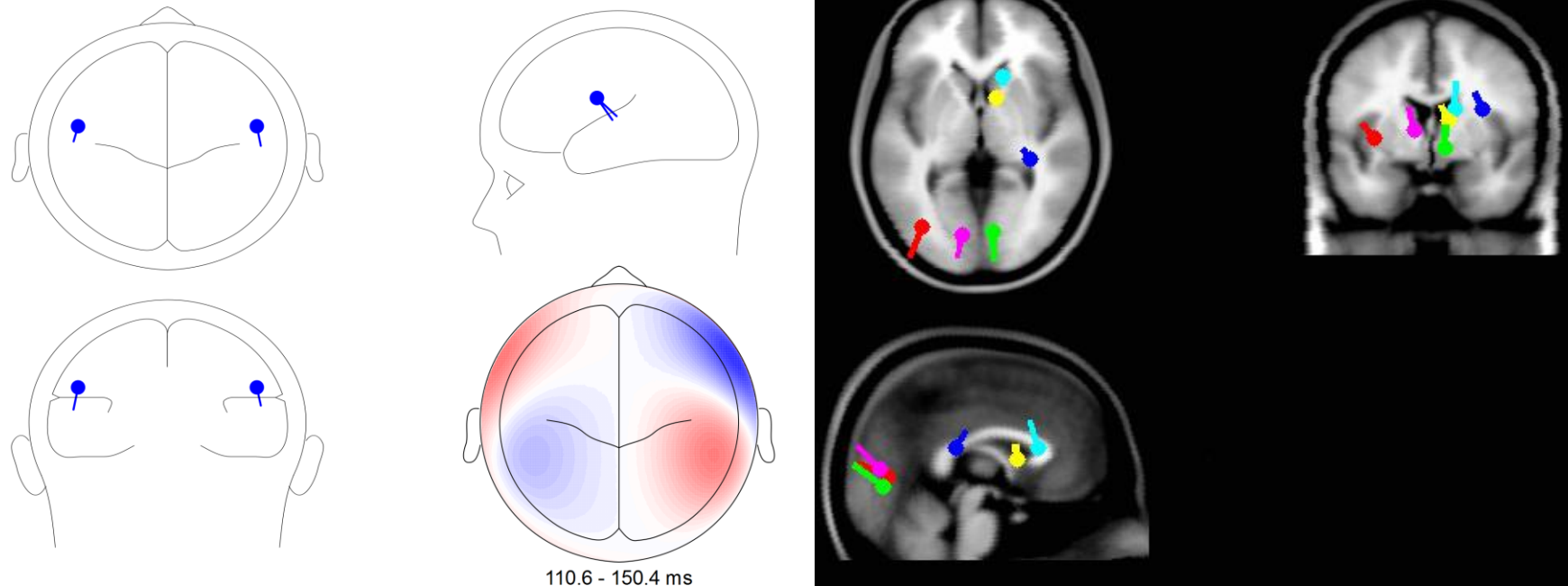
Distributed Sources

1. Assume sources are everywhere (e.g. distributed across the whole cortex)
2. Find the distribution of source strengths that explains the data...
3. ...AND fulfils other constraints



Hypothesis Testing - Dipole Fitting

Explicit assumptions about the number of **focal sources (dipoles)** are tested by fitting dipole models to the data. The common criterion for the selection of models is the **goodness-of-fit**.



It can be hard to choose the appropriate number of dipoles – a priori knowledge is required. Solutions for several/many dipoles can get stuck in local minima, and may not be robust to noise.

“Spatial Filters”: Beamformers

Assumptions:

- All sources captured in data covariance matrix C (signal and noise)
- We are interested in one source i in many sources

Aim:

Design a spatial filter $\mathbf{w}_i$ which projects maximally on the source of interest and minimally on noise sources.

Project on source of interest: $\mathbf{w}_i^T \mathbf{f}_i$

Suppress noise: $\min(\mathbf{w}_i^T C \mathbf{w}_i)$

$$\mathbf{w}_i = \frac{\mathbf{f}_i^T C^{-1}}{\mathbf{f}_i^T C^{-1} \mathbf{f}_i}$$

Linearly-Constrained
Minimum-Variance
(LCMV) Beamformer

Van Veen et al., 1997, <https://pubmed.ncbi.nlm.nih.gov/9282479/>

Create and apply these spatial filters vertex-by-vertex (dipole-by-dipole) and plot the distribution (possibly normalised by noise variance).

Spatial filters can also produce time courses for every source.

But note: The “spatial filter” interpretation applies to all linear methods, including MNE-type methods.

Minimum Norm Estimation Of Distributed Sources

$$\mathbf{L}\mathbf{s} = \mathbf{d} \Rightarrow \|\mathbf{L}\mathbf{s} - \mathbf{d}\|^2 = 0$$

(ignore noise for now)
subject to constraint

$$\|\mathbf{s}\|_2 = \min$$

yields the Minimum-Norm Least-Squares solution (“L2”)

$$\hat{\mathbf{s}} = \mathbf{G}_{MN}\mathbf{d}$$

with

$$\mathbf{G}_{MN} = \mathbf{L}^T(\mathbf{L}\mathbf{L}^T)^{-1}$$

But this is the result of mathematical desperation, and not based on physiology or what we want to know (e.g. localisation of multiple sources).

Noise and Regularisation in EEG/MEG Source Estimation

Noise and Regularization

Explaining the data 100% may not be desirable – some of the measured activity is not produced by sources in the model.

Explaining noise may require larger amplitudes in source space than the signal of interest:

Overfitting may seriously distort the solution
("variance amplification" in statistics/regression).

“Whitening” and Choice of Regularisation Parameter

Whitened data have a noise covariance that is the identity matrix – i.e. noise is “white” (uncorrelated) noise.

$$\mathbf{G}_{MN} = \mathbf{L}^T (\mathbf{L}\mathbf{L}^T + \lambda \mathbf{C}^{-1})^{-1}$$

can also be written as

$$\mathbf{G}_{\widetilde{MN}} = \widetilde{\mathbf{L}}^T (\widetilde{\mathbf{L}}\widetilde{\mathbf{L}}^T + \lambda \mathbf{I})^{-1}$$

where $\widetilde{\mathbf{L}}$ is the “whitened” leadfield $\mathbf{C}^{-1/2}\mathbf{L}$, and scaled such that $\text{trace}(\widetilde{\mathbf{L}}\widetilde{\mathbf{L}}^T) = \text{trace}(\mathbf{I})$.

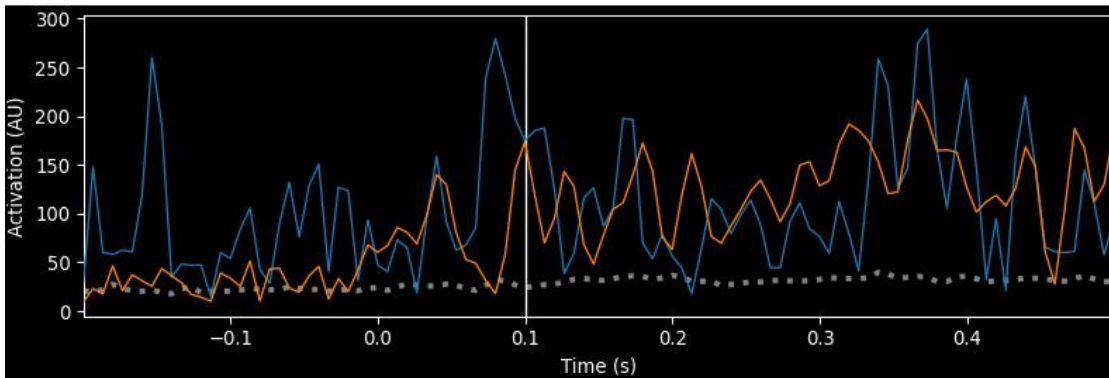
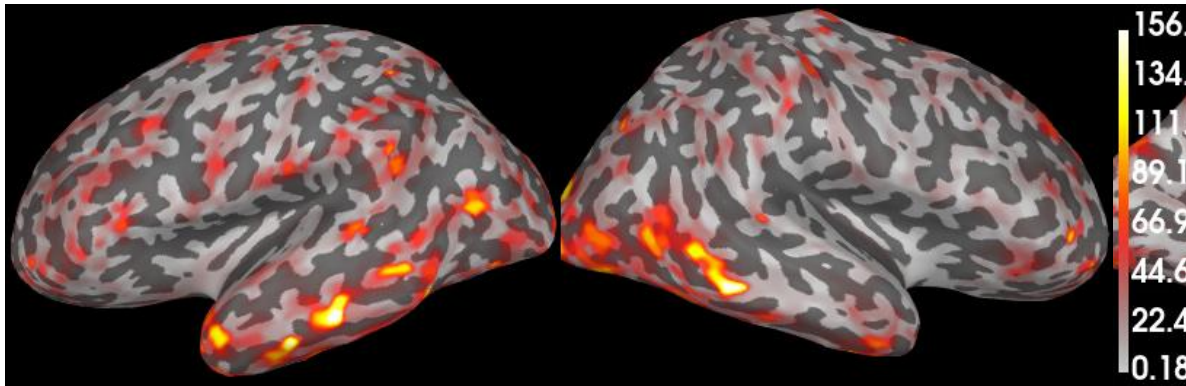
$\widetilde{\mathbf{L}}$ and λ can now be interpreted in terms of signal-to-noise ratios.

A reasonable choice for λ is then the approximate SNR of the data (e.g. in MNE software) –

usually heuristically chosen to be 3 (evoked) or 1 (raw/continuous).

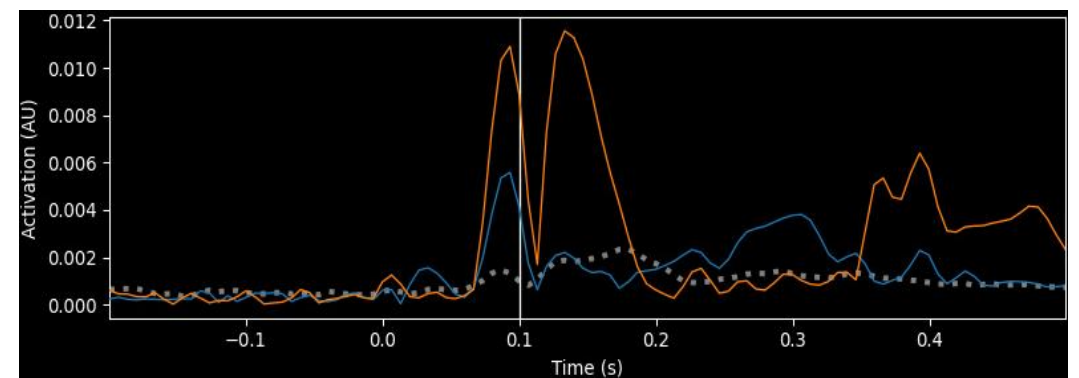
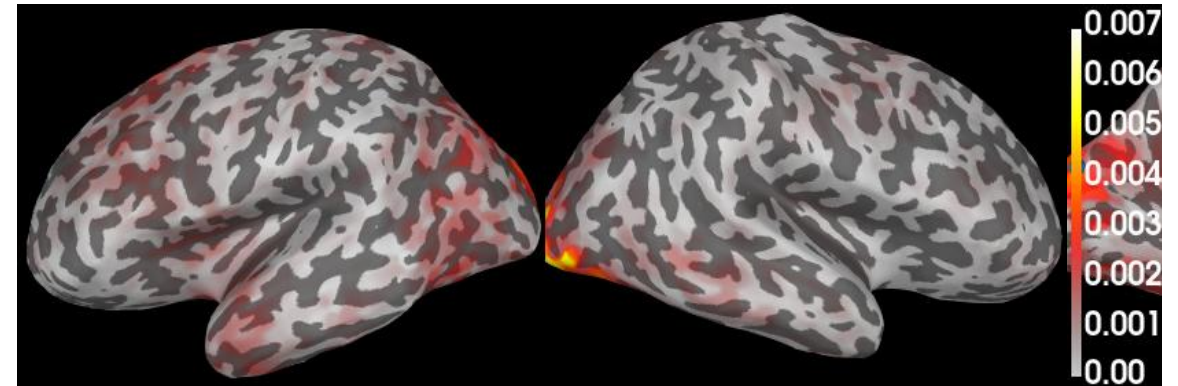
The Effect of Regularisation ~ Over- and Under-Fitting

Over-fitting
“Under-smoothing”
SNR=300



var=95.2%

Under-fitting
Over-smoothing
SNR=0.03



var=0.4%



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Thank you